

DSCI510 FINAL PROJECT

The Analysis of U.S. Economic Stress and Social Media Seeking (2019–2024)

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INTRODUCTION



1. Research Objective

To examine how the COVID-19 **economic shock** influenced individuals' behavioral preferences on **social media** from **2019 to 2024**, specifically whether people tend to shift toward **job-seeking (functional usage)** or **entertainment (recreational usage)** under rising unemployment.



2. Research Questions

- **Q1:** What are the trends in unemployment before, during, and after COVID-19?
- **Q2:** Does higher unemployment lead to increased functional social media usage?
- **Q3:** Does higher unemployment lead to increased recreational social media usage?



3. Analytical Focus

- Unemployment trends across high-stress states
- Functional vs. recreational social media responses to economic stress

Data Sources & Approach

Data Source	Description	Processing Approach
BLS Unemployment Data (3600 data points) https://download.bls.gov/pub/time.series/la/	Monthly unemployment rate by state (2019–2024). Used for measuring economic stress	Retrieved via BLS API ; cleaned and structured as time-series data
Google Trends (Functional Social media behaviour: 2160 data points) https://trends.google.com/trends/explore	Search interest index (0–100) for Job-seeking Keywords Category : “LinkedIn”, “Job”, “Hiring” to capture job-seeking behavior	Collected via pytrends ; keywords combined and averaged to construct a recreational index
Google Trends (Recreational Social media behaviour: 2160 data points) https://trends.google.com/trends/explore	Search interest index (0–100) for Entertainment Keywords Category : “TikTok”, “Netflix”, “Instagram” to capture entertainment behavior	Collected via pytrends ; keywords combined and averaged to construct a recreational index

Notes:

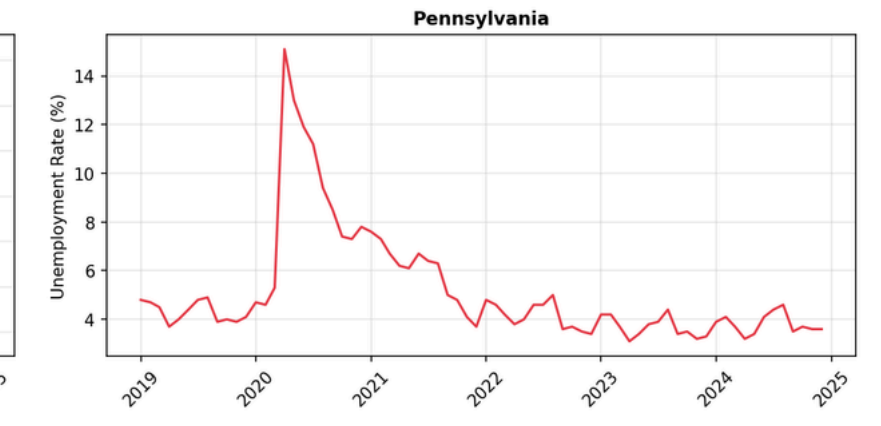
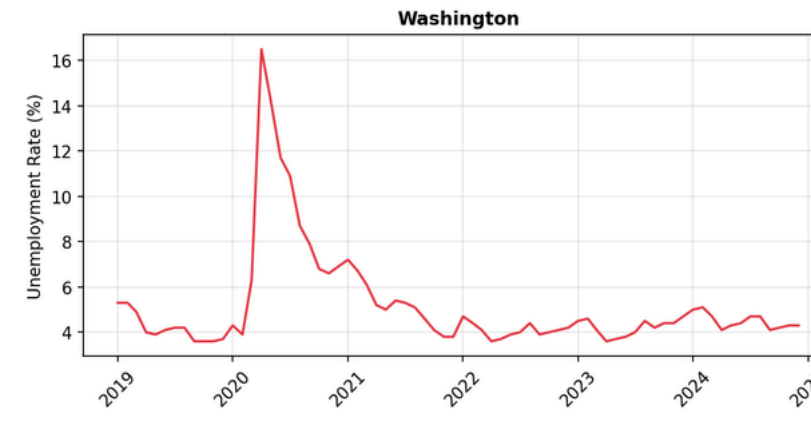
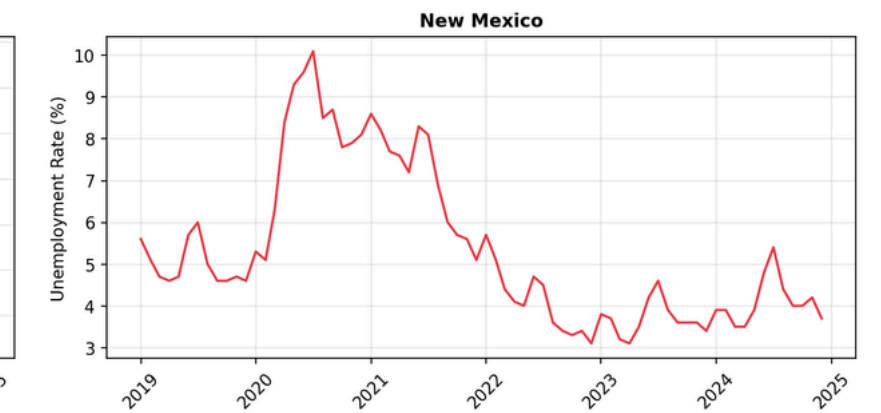
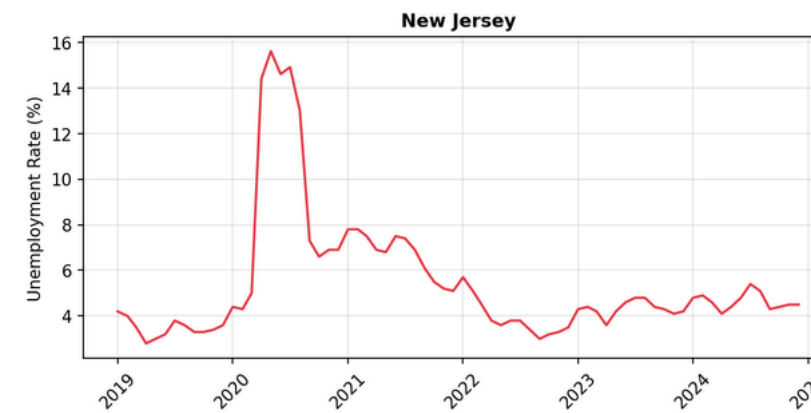
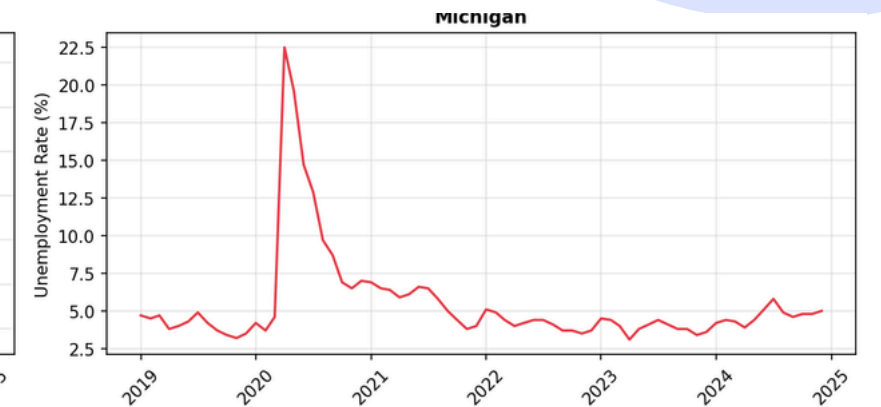
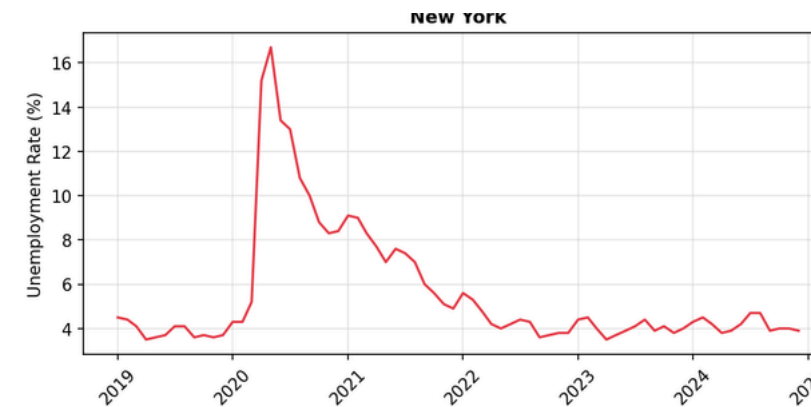
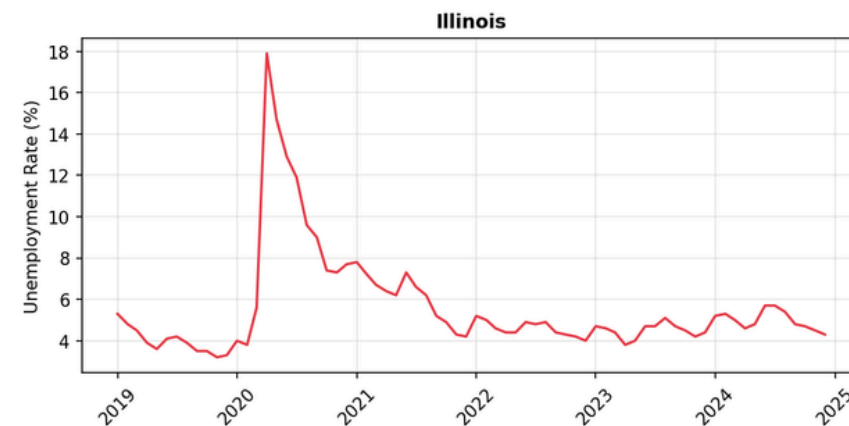
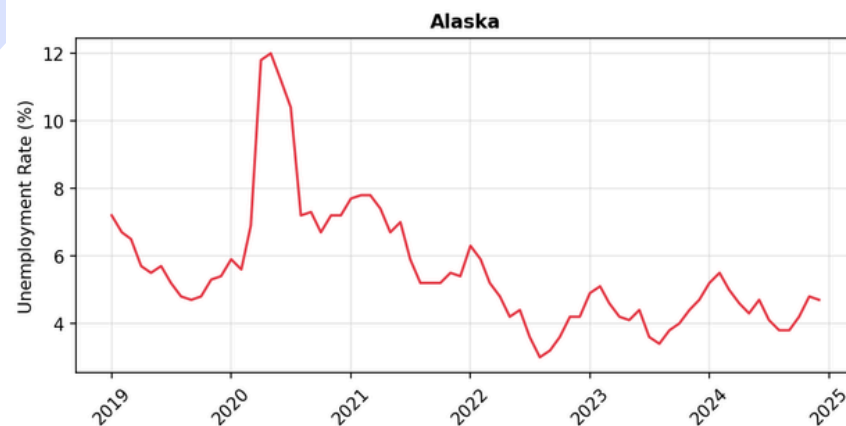
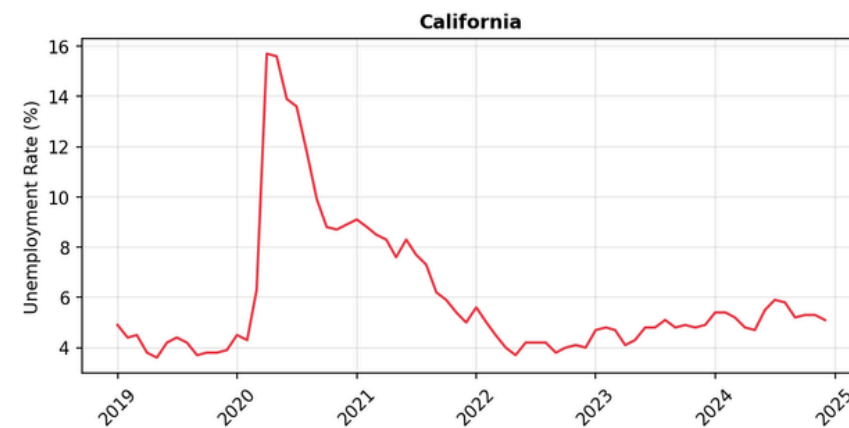
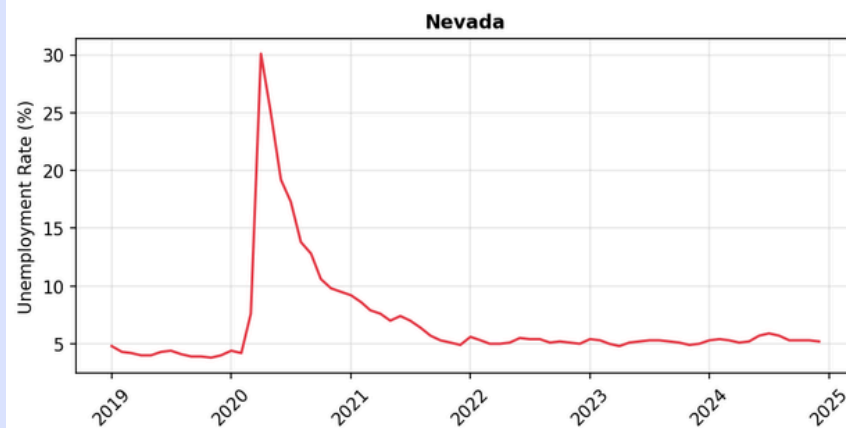
Search interest = how popular a keyword is over time (0–100), not how many people search it, not actual search counts
Pytrends is a Python library for accessing Google Trends data, which is used to retrieve keyword-level search indices

Result 1-The unemployment trends

Unemployment peaked in 2020 and declined afterward:

- Top 10 high-unemployment states(2019-2024) identified:
Nevada, California, Alaska, Illinois, New York, Michigan, New Jersey, New Mexico, Washington, Pennsylvania
- **Sharp increase during COVID-19 (2020)**
- **Gradual recovery** from 2021 to 2024
- Persistent variation across states

Monthly Unemployment Rate (2019-2024)
Top 10 Highest-Unemployment States



New York

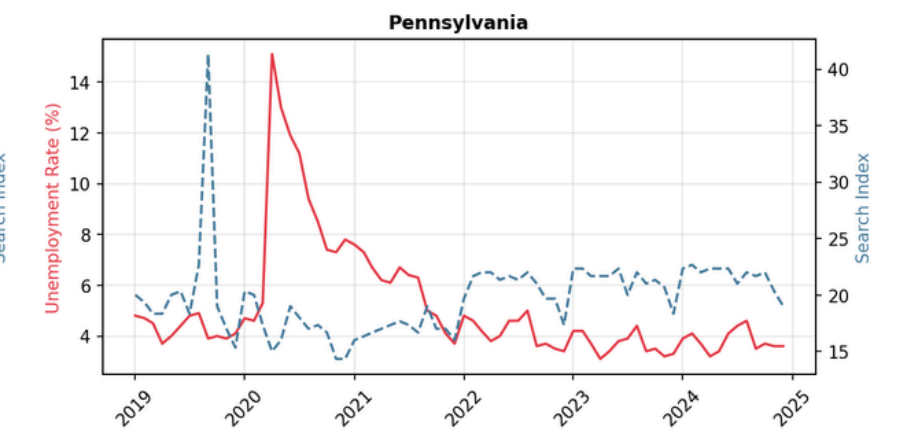
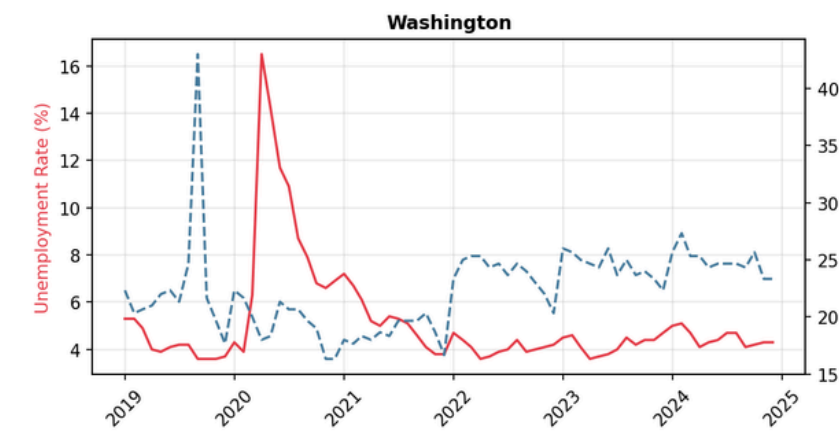
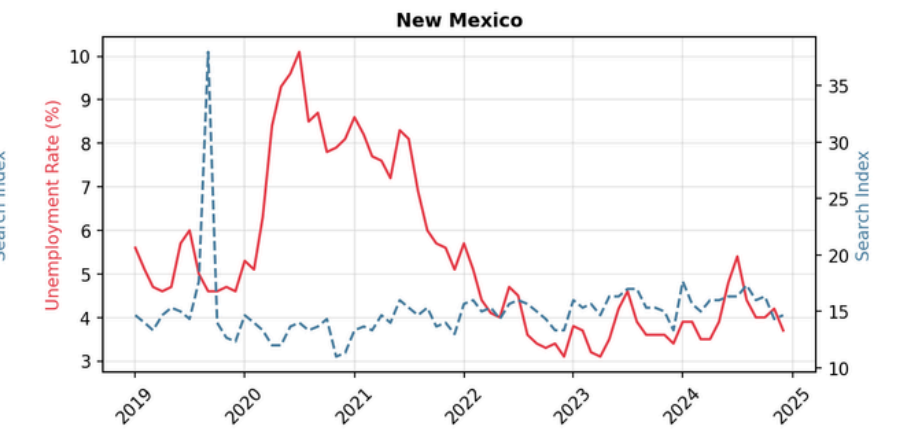
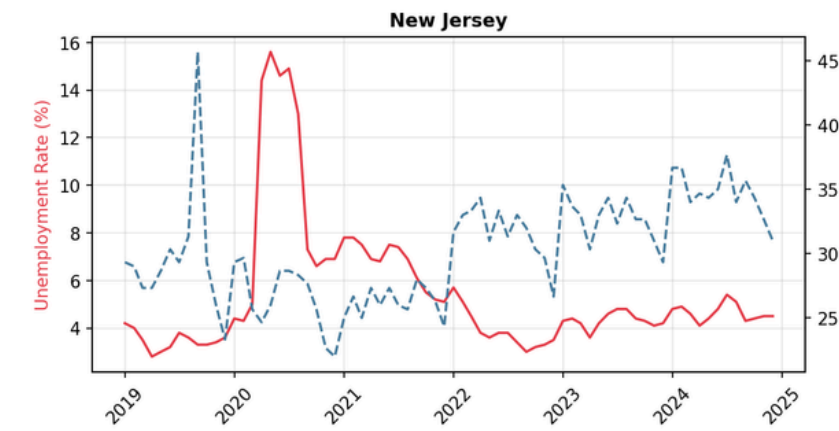
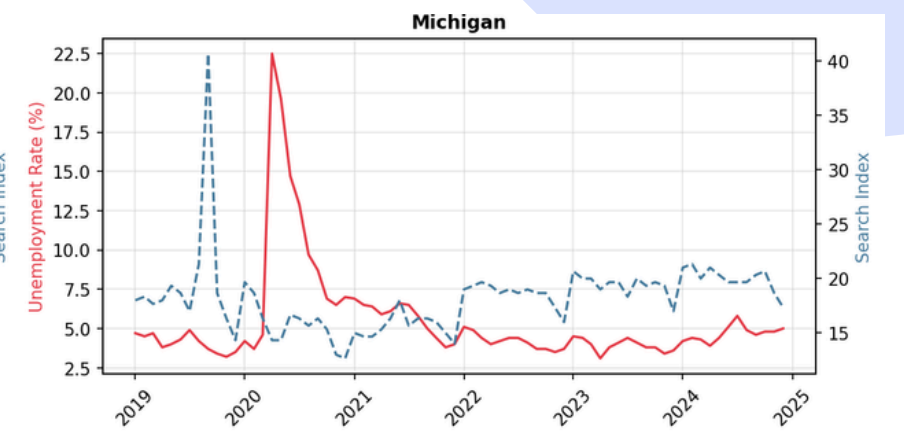
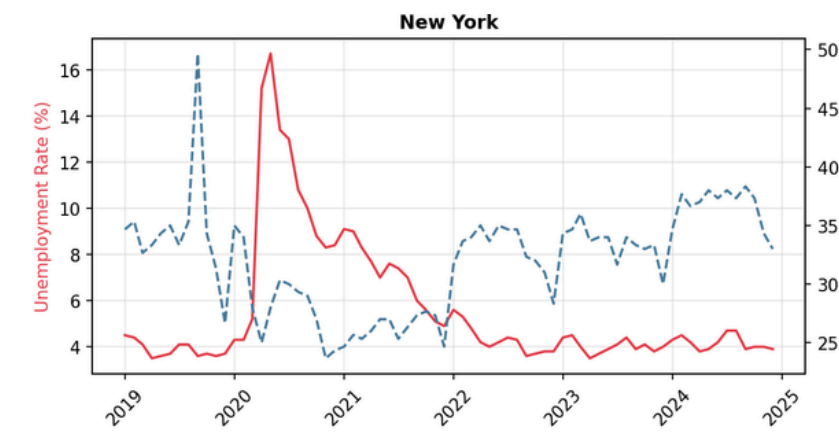
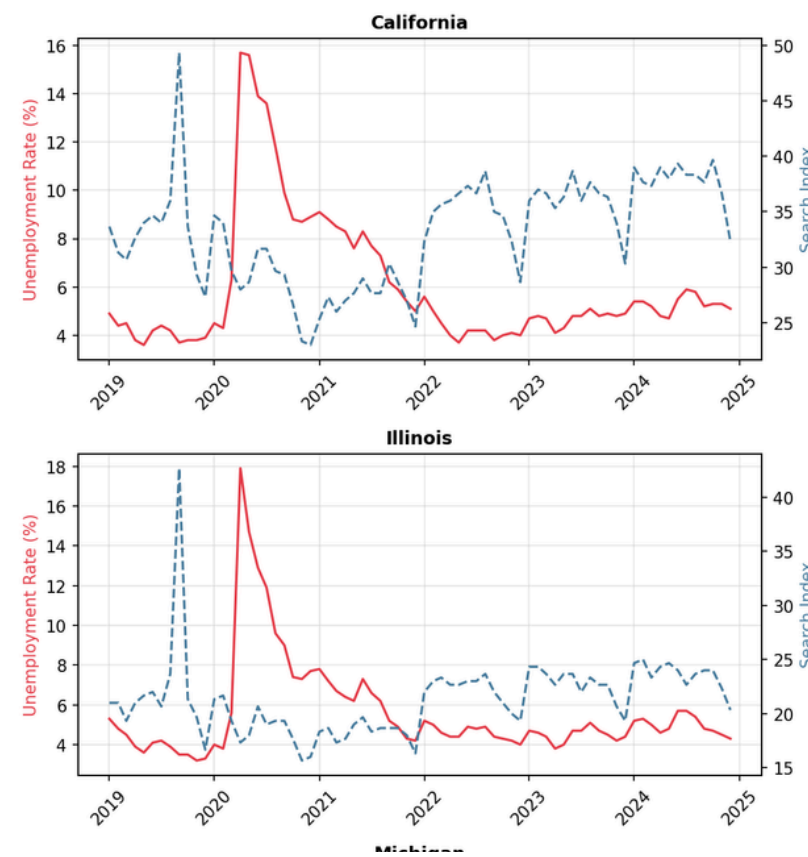
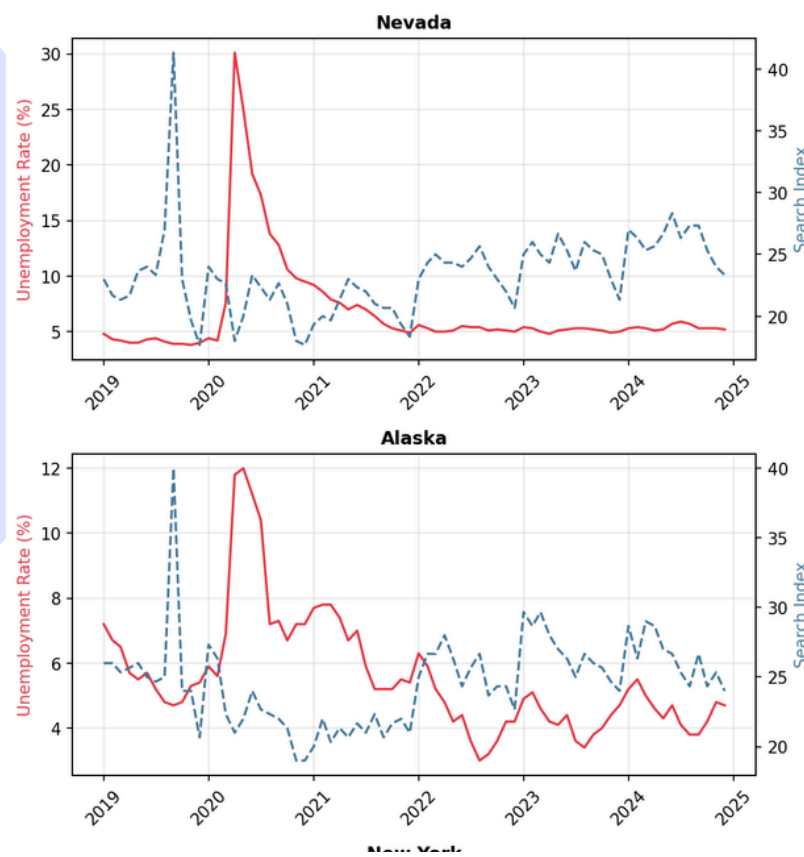
Michigan

Results 2: Functional Social Media

Functional search interest increases before unemployment peaks

- **Drops or remains low** at peak unemployment
- **Increases** after unemployment starts to **decline**
- Suggests lagged job-seeking behavior
- Indicates **delayed response to economic shock**

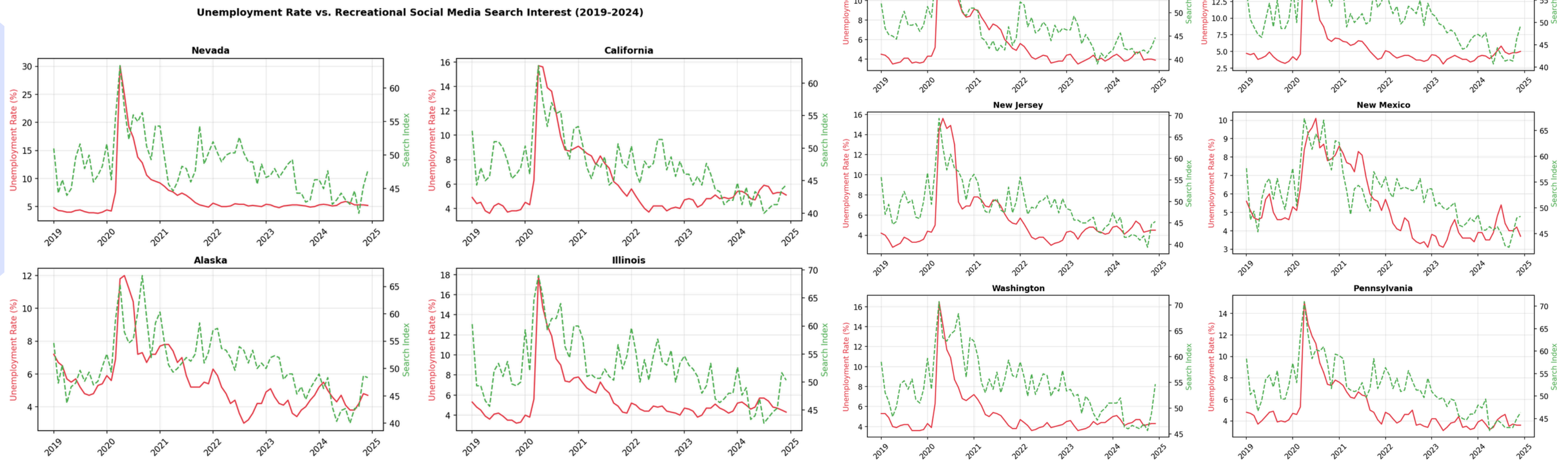
Unemployment Rate vs. Functional Social Media Search Interest (2019-2024)



Results 3: Recreational Social Media

Recreational search interest rises during the shock but declines as unemployment recovers

- **Increases** during the 2020 unemployment spike
- **Gradually declines** as unemployment decreases
- Returns closer to pre-pandemic levels
- Suggests **a short-term response to economic shock**

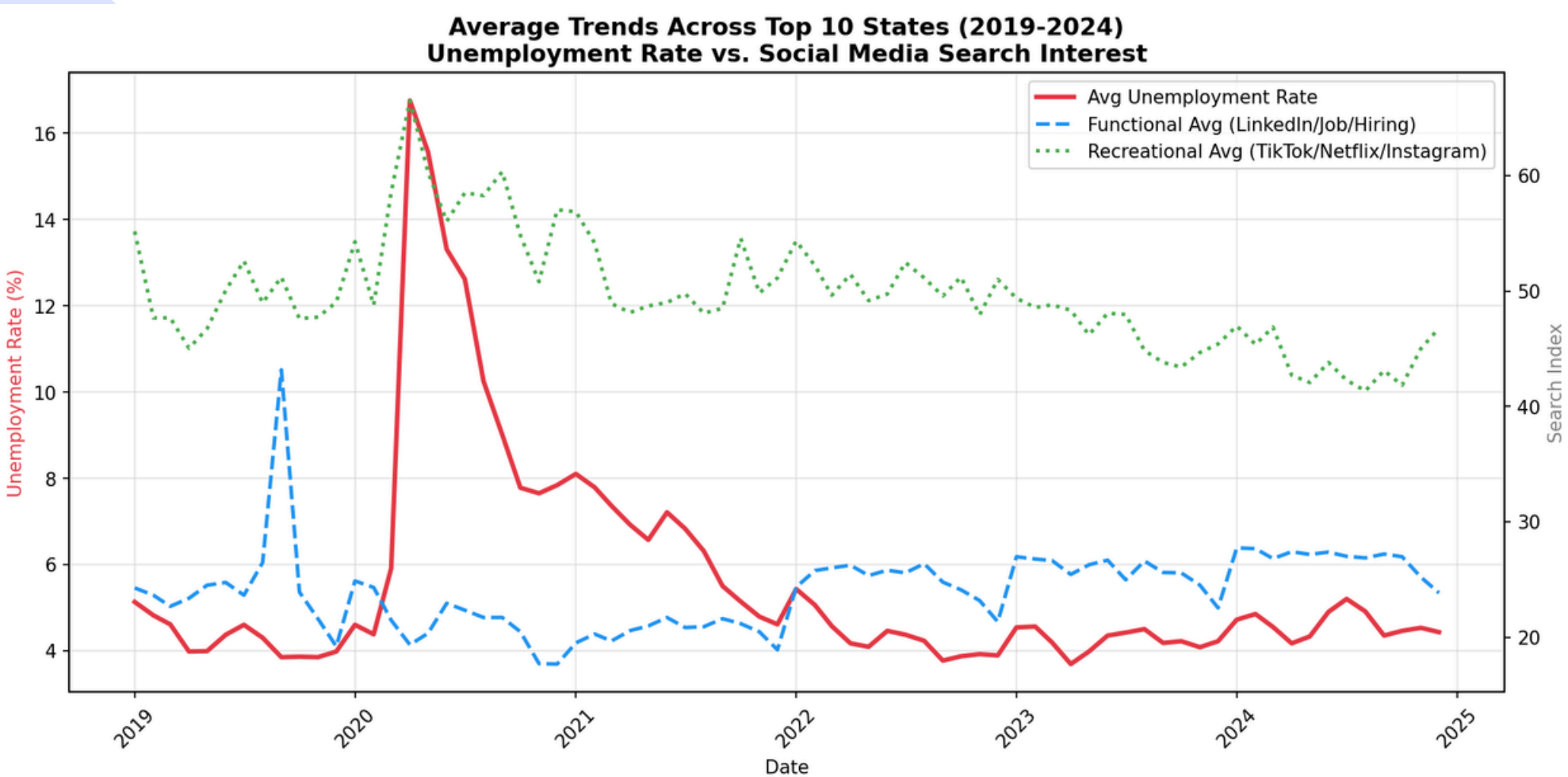


Results 4-Overall Trends

Functional and recreational Social Media behaviors show different long-term patterns

- Job-Seeking Social media behavior increases around the unemployment peak and after the unemployment peak, suggesting a growing response to economic stress.
- Entertainment Social media behavior rises during the unemployment spike and declines afterward (immediate response)

Higher unemployment → more recreational social media (Moderate positive correlation: $r = 0.661$, $p < 0.05$)
Higher unemployment → less job-seeking behavior (Weak positive correlation: $r = -0.400$, $p < 0.05$).



--- Overall Summary ---

Avg Functional correlation: $r = -0.400$

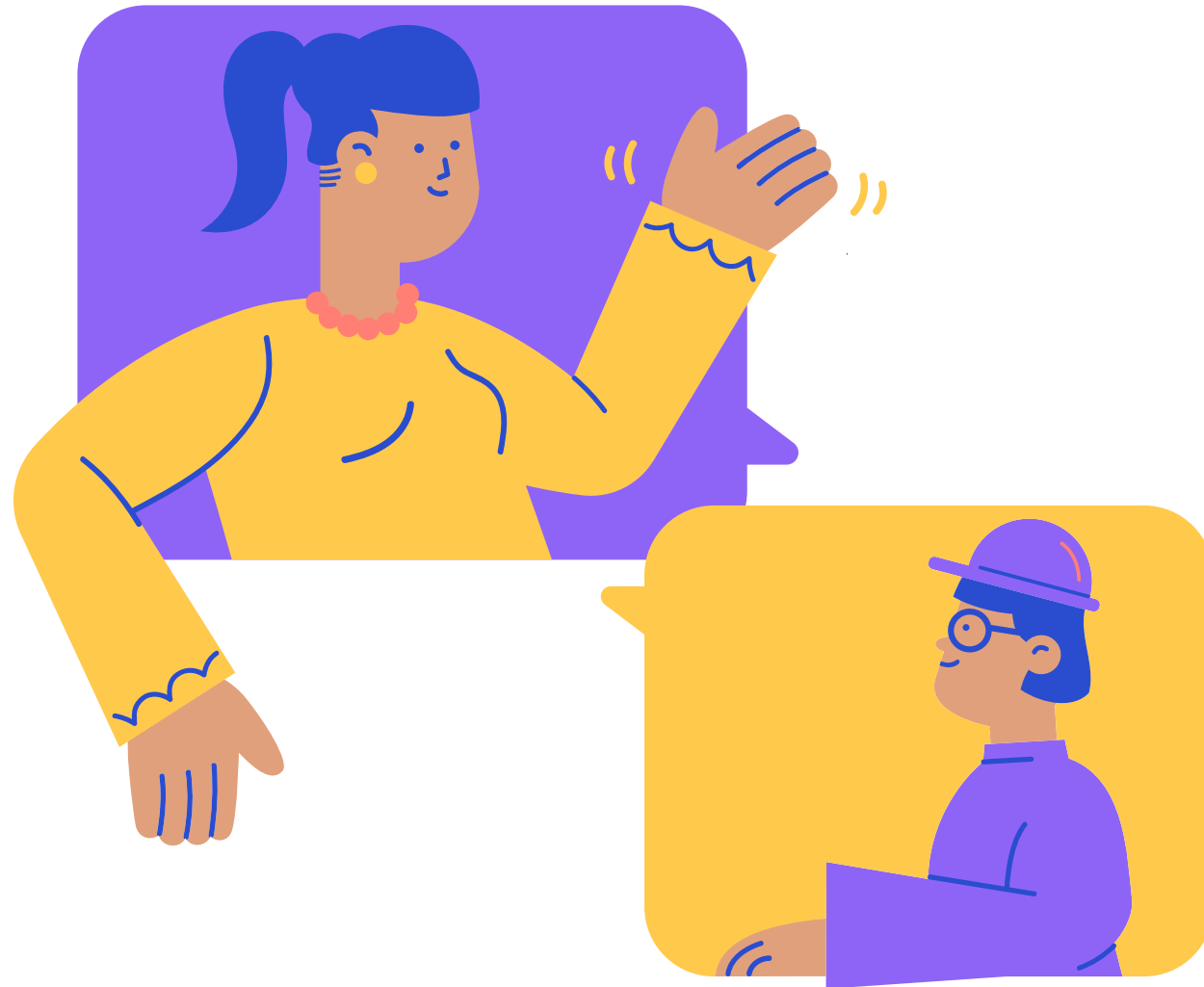
Avg Recreational correlation: $r = 0.661$

Correlation results table:

state	func_corr	func_pvalue	rec_corr	rec_pvalue
Alaska	-0.410	0.0003	0.580	0.0
California	-0.504	0.0000	0.632	0.0
Illinois	-0.365	0.0016	0.676	0.0
Michigan	-0.350	0.0026	0.557	0.0
Nevada	-0.344	0.0031	0.678	0.0
New Jersey	-0.363	0.0017	0.676	0.0
New Mexico	-0.270	0.0219	0.709	0.0
New York	-0.598	0.0000	0.663	0.0
Pennsylvania	-0.417	0.0003	0.739	0.0
Washington	-0.379	0.0010	0.704	0.0

All statistically significant

Challenge



Data Collection Challenges

- Google Trends (pytrends) may trigger **Error 429 (Too Many Requests)**
- Caused by sending frequent automated requests
- Resolved by adding time delays between requests
- **BLS API may be unstable**, returning inconsistent data sizes



Data Limitations

- Search interest is a relative index (0–100), not actual search counts
- Limits direct comparison across different keywords

Thanks for Listening

